

Beyond Token Serialization: Recurrent Latent Relaxation and In-Context Adaptation in BDH-CQ

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Abstract

Standard autoregressive Large Language Models (LLMs) rely on verbal Chain-of-Thought (CoT) prompting to execute multi-step reasoning. We argue that token serialization is an architectural crutch for the fixed computational depth of standard Transformers rather than a biological necessity. Pathway’s Dragon Hatchling (BDH) architecture family, specifically BDH-CQ, introduces an alternative: few-shot demonstrations are assimilated into recurrent associative memory during the forward pass, and reasoning is performed via iterative latent relaxation. This technical briefing formalizes BDH-CQ mechanics, examines verified benchmark results on ARC-AGI-1 and Sudoku Extreme, and evaluates the non-trivial trade-off between compute efficiency and intermediate token-level interpretability.

1 The Token Serialization Bottleneck

Autoregressive Transformers have static layer depth L , strictly bounding the compute executed per forward pass. Chain-of-Thought (CoT) circumvents this constraint by externalizing intermediate steps as emitted natural-language tokens, feeding each symbol back into the attention window so subsequent passes borrow computational depth from sequence length. However, token serialization introduces three fundamental structural inefficiencies:

1. **Linear Memory Expansion:** State footprint scales as $\mathcal{O}(T)$ in token sequence length, inflating Key-Value (KV) cache memory and saturating high-bandwidth memory (HBM).
2. **Token Compute Overhead:** Full autoregressive projection is paid for conversational syntax, punctuation, and linguistic filler that carry negligible logical entropy.
3. **Verbosity-Bound Cost:** Inference FLOPs scale with natural language verbosity rather than intrinsic problem complexity.

2 Formal Mechanics of BDH-CQ

BDH-CQ eliminates intermediate token generation by moving reasoning into continuous latent dynamics [1, 2]:

1. Forward-Pass Demonstration Assimilation. Unlike test-time optimization frameworks that execute gradient descent across demonstration samples, BDH-CQ updates an associative context state S_t strictly during the inference forward pass:

$$S_t = U_\theta(S_{t-1}, D_t) \quad (1)$$

where $D_t = (x_t, y_t)$ is an observed input-output demonstration tuple and U_θ is a learned, frozen associative update operator. Context is compressed into fixed-rank associative memory using linear attention and non-negative ReLU low-rank projections, maintaining invariant $\mathcal{O}(1)$ working memory complexity [1].

2. Recurrent Latent Relaxation. Queries are resolved by repeatedly cycling a continuous hidden vector $h \in \mathbb{R}^d$ through a recurrent operator across depth $t \in \{1, \dots, T\}$ without textual token generation:

$$h_{t+1} = \text{LayerNorm}(h_t + \alpha \cdot \text{ReLU}(Wh_t - b)) \quad (2)$$

The recurrent operator W and bias b define a dynamical system whose attractor basins correspond to constraint satisfaction, constrained by biologically inspired non-negative activation sparsity ($\sim 5\%$ active units) [2]. Sequence length remains invariant ($\mathcal{O}(1)$).

3 Architectural Landscape Comparison

Table 1 contrasts BDH-CQ against token-based CoT and optimization solvers on the ARC-AGI benchmark [3].

Dimension	Standard Dense Transformer + CoT	Test-Time Training (TTT) / HRM / TRM	BDH-CQ (Pathway Research)
Sequence Length	$\mathcal{O}(T)$ (expands per reasoning token)	$\mathcal{O}(1)$ (fixed prompt structure)	$\mathcal{O}(1)$ (invariant continuous state)
KV-Cache Scaling	$\mathcal{O}(T)$ (linearly growing memory footprint)	$\mathcal{O}(1)$ (no token accumulation)	$\mathcal{O}(1)$ (flat constant working buffer)
Test-Time Adaptation	Appended prompt context tokens	Test-time backpropagation / weight updates	Forward-pass associative state updates (U_θ)
Compute Mechanism	Autoregressive softmax sampling per step	Multi-epoch backward-pass search on prompt	Low-rank recurrent forward relaxation
Observability	High (human-readable English tokens)	Low (parameter distribution shift)	Low (continuous vector trajectory in $\mathbb{R}^d$)

Table 1: Architectural comparison of test-time reasoning paradigms.

4 Empirical Evidence & Pareto Dynamics

Verified benchmarks establish that recurrent latent computation shifts the “intelligence-per-dollar” Pareto frontier:

- **ARC-AGI-1 Benchmark [3]:** A 150M parameter BDH-CQ model achieved **29.5% pass@2** at a computed inference cost of **\$0.0007 per task** (measured on NVIDIA H200 hardware under a \$3.00/hr compute standard) [1]. By comparison, frontier autoregressive models such as GPT-5.6 Luna Low scored 34.2% at \$0.0077 per task—an $11\times$ cost premium for a 4.7 percentage point accuracy margin.
- **Sudoku Extreme Evaluation:** Across $\sim 250,000$ extreme puzzles, Pathway BDH attained **97.4% accuracy** without emitting verbal tokens, executing backtracking routines, or relying on external symbolic verifiers [2], where dense autoregressive LLMs suffered from compounding token hallucination.

5 Scientific Caveats & Honest Limitations

Recurrent latent relaxation exhibits concrete trade-offs and structural failure modes documented in controlled interventions:

1. **The Observability Gap:** CoT generates an auditable text scratchpad. In BDH-CQ, intermediate trajectories occur in continuous latent space; diagnostic failure analysis requires probing projection heads or sparse autoencoders (SAEs).
2. **Non-Convex Attractor Trapping:** Recurrent relaxation is an energy minimization process. On deceptive constraint puzzles, h_t can stall in shallow local minima without the explicit backtracking pathways available to rejection-sampled token models.
3. **Compositional Property Binding Failure:** Controlled interventions in [1] reveal catastrophic degradation on multi-attribute compositional binding (e.g., scoring **0/72 on color-swap composition**), exposing a vulnerability in disentangling relational visual features.
4. **Domain Bounds:** Verified advantages are proven on discrete spatial constraint grids (ARC-AGI, Sudoku); generalization to open-ended dialogue and external tool use remains unestablished.

References

- [1] B. Engdahl, A. Kosowski, J. Chorowski, Z. Stamirowska, P. Uznański, R. Zhong, et al. *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888, 2026.
- [2] Pathway Research. *BDH (Dragon Hatchling): A Brain-Inspired Post-Transformer Architecture*. Technical Report, 2025/2026.
- [3] F. Chollet. *On the Measure of Intelligence*. arXiv:1911.01547, 2019.